1 **(a)** A positively charged metal sphere A is brought close to a negatively charged metal sphere B as shown in Figure 1.1. The charge on metal sphere B is twice that of the charge on metal sphere A.

On Figure 1.1, draw lines to represent the electric field around the spheres.



Figure 1.1 [2]

(b) Point P is at a distance *x* from the centre of sphere A along the line joining the centres of the two spheres as shown in Figure 1.2. The radius of A and B is 15 mm and the distance between the centres of the spheres is 80 mm.

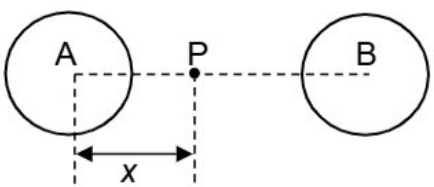


Figure 1.2

The variation with *x* of the electric potential *V* at P is shown in Figure 1.3.

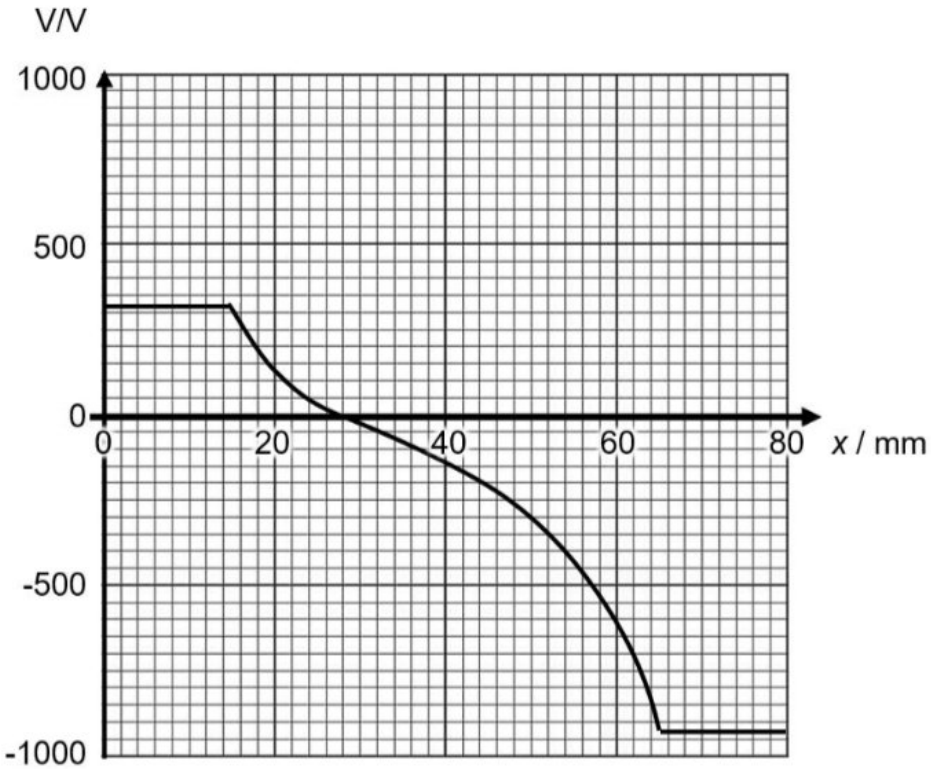


Figure 1.3

(i) Determine the magnitude of the electric field strength at P where x = 40 mm.

electric field strength = ... V m^{-1} [2]

(ii) An electron is released from rest at point P where x = 40 mm.

1. Describe and explain the motion of the electron as it travels 20 mm along the line joining the centres of the spheres.

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..... [3]

2. Determine the speed of the electron when it has travelled 20 mm along the line joining the centres of the spheres.

speed = ……m s^{-1} [4]

[Total: 11]